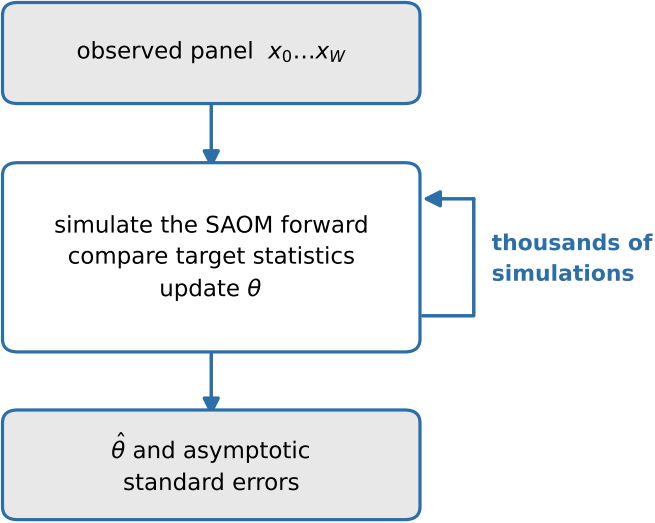


Two ways to fit the same model by simulating from it

RSiena (siena07)

the loop is around your dataset

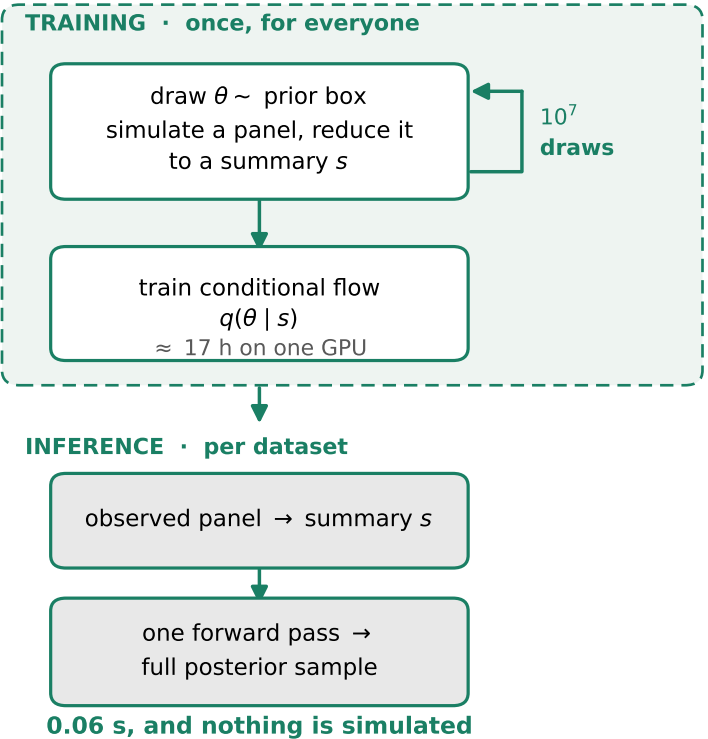


every dataset pays this again, in full

219 s for one three-wave panel, $n = 129$

Amortized neural estimator

the loop is in training, over the prior

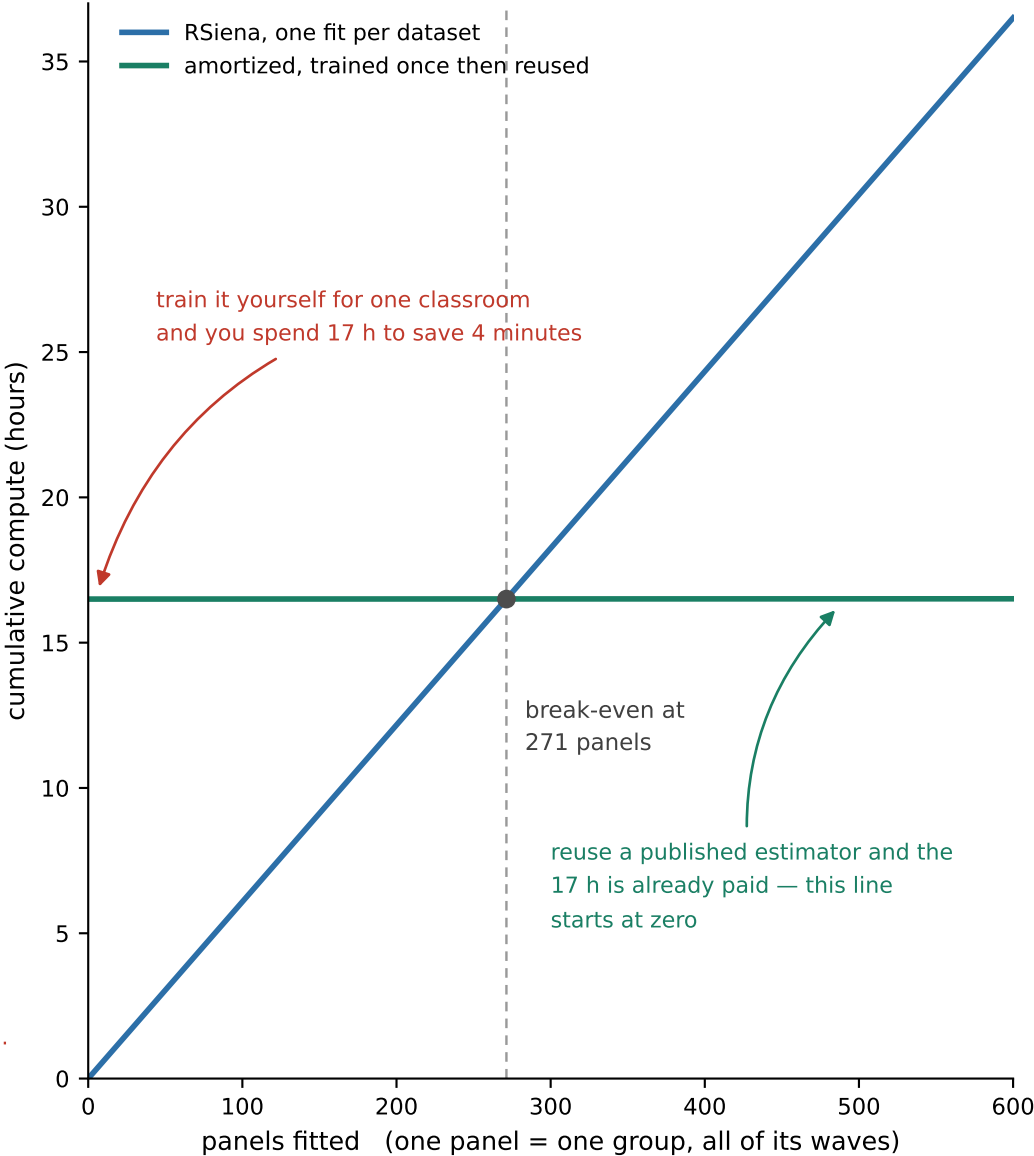


What you give up

RSiena: the specification is chosen when you run it — any effect set, any n , any number of waves.

Amortized: effect set, n range and priors are fixed at training time. Outside them the posterior does not get noisy — it gets confident and wrong.

Amortization is a claim about the second fit, not the first



A **wave** is one observation of the network; a **period** is the interval between two consecutive waves, and a W -wave panel has $W - 1$ of them; a **panel** is one group observed at all of its waves — one dataset.

Fit times are the measured ones for the three-wave Glasgow panel ($n = 129$); the 17 h is 5 h 34 m to generate 10^7 panels plus 10 h 56 m to train, for the M6 network estimator. Both are single runs on one machine, not benchmarks.